

Khushi Patel

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EDUCATION

Northeastern University, Khoury College of Computer Sciences
B.S. Computer Science and Philosophy

Sept 2021 - May 2025
Boston, MA

SUMMARY

Generative AI Engineer with experience building, fine-tuning, and deploying LLMs and multimodal AI systems. Skilled in RAG pipelines, prompt engineering, and productionizing scalable AI solutions for enterprise and public-sector applications.

TECHNOLOGIES

Languages: Python, Javascript, Typescript, Java, R, HTML, CSS, SQL, [React.js](#)

AI Tools/Skills: AWS Bedrock (Claude, Llama, Mistral), Vector Databases (e.g., Pinecone), LangChain, RAG Architecture, LLM Evaluation, Context Window Management

EXPERIENCE

Commonwealth of Massachusetts

Sept 2025- Present

AI Developer

Remote

- Leading development of **5+ internal and cross-secretariat AI products**, acting as primary technical lead and managing a team of 3+ interns
- Launched **GENIE**, Massachusetts' first Generative AI safe and secure chatbot; responsible for full lifecycle: architecture, deployment, maintenance, and rollout to **44,000 state employees**
- Optimized **GENIE latency by 50%** through redesigning RAG backend architecture; implemented a frontend prompt enhancement feature for GENIE
- Developed a **generalized RAG chatbot framework** for replication across TSS, enabling AI capabilities for multiple business functions

Commonwealth of Massachusetts

Jan 2025- Aug 2025

Artificial Intelligence Intern

Remote

- Enhanced GENIE platform with **secure document upload, multi-chat playground, and customizable AI tasks**, improving usability and adoption; developed backend prompt enhancer improving prompt accuracy by 60%
- Fine-tuned backend LLM pipelines for prompt optimization and integrated frontend React.js enhancements for superior UX

Burnes Center for Social Change

July 2024 – Dec 2024

Generative AI Product Developer

Boston, MA

- Led full-cycle development of an **AI-powered knowledge base for the Executive Office of Economic Development (EOED)**, enabling small businesses to seamlessly access 50+ grants/programs from the state of MA
- Designed and implemented a **Retrieval Augmented Generation (RAG)** backend architecture; designed and deployed **RESTful APIs using AWS API Gateway and Lambda**, integrating with microservices for scalable backend processing
- Built an intuitive frontend allowing users to filter grants/programs based on specific parameters, seamlessly integrating with the backend to deliver highly accurate, AI-driven grant recommendations via chatbot
- Optimized EOED response times, reducing employee turnaround from **2–14 days to just 1–5 minutes**, significantly improving operational efficiency and accessibility

Amazon Robotics

July 2023 – Dec 2023

Software Technical Program Manager

North Reading, MA

- Working with AWS services/Python to develop script that performs real-time analysis of robotic arm collisions, **allowing 15+ teams** to quickly identify root cause of collisions and **increasing previous analysis speed by 2x**
- Responsible for software deployment automation of Python scripts, streamlining processes including software updates, network configuration, etc
- Managed Robotic Orchestration and Controls team, **increased software updates deployment by 2x** and established scrum process from scratch, setting up regular meetings and sprint organization
- Assisted SQA team in operating robotic workcells, creating user manuals and training new hires in workcell operation

